Lagrange's Theorem

If G has a subgroup H and $|G|$ is finite then $|H|$ divides $|G|$.

Converse of Lagranges Theorem is the question: If $d \mid |G|$, does G have a subgroup H order d?

Maybe, but not always

eg

~~Prop 6.15~~ Consider A_4, alternating group on 4 elements, $A_4 = \{$even permutations from $S_4\}$

$$|A_4| = \frac{|S_4|}{2} = \frac{4!}{2} = 12$$

and A_4 has no subgroup of order 6

$\forall g \in A_4$ $\quad gH = Hg$

$\Rightarrow gHg^{-1} = H$

Nor $gHg^{-1} = \{ghg^{-1} : h \in H\} = H$

eg $(1\ 2\ 3) \in A_4$

$|A_4| = 12$

$|A_4 \times H|$

~~Q1~~ So ~~suppose~~ $a, b \in G$,

and $|a| = 5$, $|b| = 7$.

Claim: $|G| \geq 35$.

How might Lagrange's Theorem apply here?

$\langle a \rangle$, $\langle b \rangle$ are both subgroups of G

and $|\langle a \rangle| = |a| = 5$

$|\langle b \rangle| = |b| = 7$.

Lagranges theorem says, if $|G|$ infinite.

then $5 \mid |G|$ and $7 \mid |G|$.

$\Rightarrow$ $5 \cdot 7 = 35$ divides $|G|$, since

$5, 7$ are coprime pair.

(from chap 2 of NT.

$[a \mid n \;\&\; b \mid n \;\&\; \underline{\gcd(a,b) = 1}]$

$\Rightarrow ab \mid n$.

<u>Note</u> $25 \mid 100$ & $5 \mid 100$

but $5 \cdot 25 = 125 \nmid 100$

$\Rightarrow |G| \geq 35$

<u>Q2)</u> If $|G| = 60 = 2^2 \cdot 3 \cdot 5$ and H is a subgroup of G. Then $|H|$ must be one of.

1, 2, 3, 6, 5, 10, 12, 15, 30, 20, 60

Q5. (a)

The left cosets of $\langle 8 \rangle$ in $\mathbb{Z}_{24}$ are.

$H = \langle 8 \rangle = \{0, 8, 16\}$. So we expect eight coset of H in $\mathbb{Z}_{24}$.

- $H = 0 + \langle 8 \rangle = 8 + \langle 8 \rangle = 16 + \langle 8 \rangle$
- $1 + \langle 8 \rangle = \{1, 9, 17\} = 9 + \langle 8 \rangle = 17 + \langle 8 \rangle$
- $2 + \langle 8 \rangle$

$3 + \langle 8 \rangle$

$4 + \langle 8 \rangle$

$5 + \langle 8 \rangle$

$6 + \langle 8 \rangle$

$7 + \langle 8 \rangle$

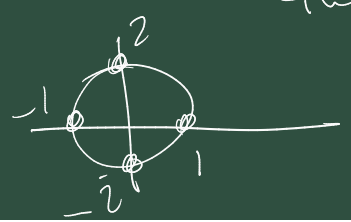
Since $\mathbb{Z}_{24}$ is abelian, the right cosets are the same.

(g) $\mathbb{T}$ in $\mathbb{C}^*$

where $\mathbb{C}^*$ = multiplicative group of non-zero complex numbers

$$\mathbb{T} = \{ z \in \mathbb{C}^* : |z| = 1 \}$$

the unit circle in complex plane



So let $z \in \mathbb{C}^*$

What is the left coset $z\mathbb{T}$?

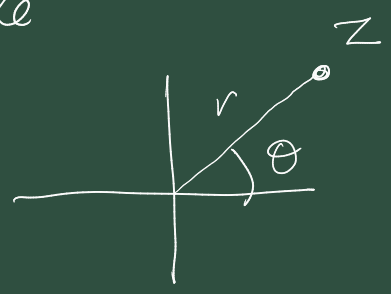
$$z\mathbb{T} = \{ zw : w \in \mathbb{T} \}.$$

But what is this? How does it relate to z?

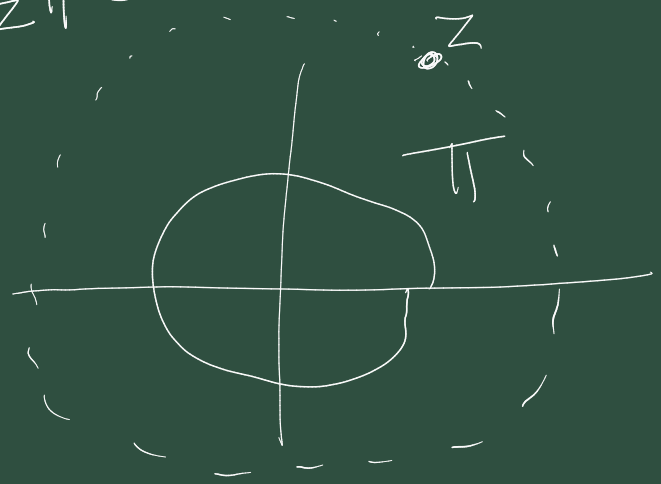
$z = a + ib$, Cartesian coords, maybe not so useful here

$= re^{i\theta}$, r = radius of z, ie $r = |z|$

θ = polar angle



$z\mathbb{T} \subset$ circle of radius $|z| = r$



Can we say something about polar coords of an element of $z\mathbb{T}$?

So let $x \in z\mathbb{T}$

so $x = zt$, for some $t \in \mathbb{T}$.

ie. $|t| = 1$

$|x| = |zt|$

$$= |z||t|$$
$$= |z|$$
$$= r$$

$\Rightarrow$ $\boxed{z\mathbb{T} \subset \text{circle of radius } r = |z|}$

Consider any $x \in \mathbb{C}^*$, with $|x| = r$

Note $x = z\dfrac{x}{z}$

and $\left|\dfrac{x}{z}\right| = \dfrac{|x|}{|z|} = \dfrac{r}{r} = 1.$

$\Rightarrow$ $x = z\dfrac{x}{z} = zt$, where $t = \dfrac{x}{z} \in \mathbb{T}$.

$\Rightarrow$ $\boxed{\text{circle of radius } r \subset z\mathbb{T}.}$

Therefore $z\mathbb{T} =$ circle of radius $|z|$

$\boxed{\text{Q6}}$ Let $G = GL_n(\mathbb{R})$ i.e. $\det \neq 0$.

$=$ group of invertible $n \times n$ matrices under multiplication.

G has the subgroup $S = SL_n(\mathbb{R})$

$= \{A \in G : \det(A) = 1\}$

Describe the left cosets of S in G.

(Continue at 5:05 pm)

Let $X \in G$ and consider the left coset

$$XS = \{ XA : A \in S \}.$$

What is XS? How does it relate to X?

Focus on determinants.

$$\det(XA) = |XA| \quad , \quad \underline{A \in S}$$

$$= |X||A| \quad , \text{ lin. alg. prop}$$

$$= |X| \quad , \text{ as } \underline{|A| = 1}$$

So let's conjecture that

$$XS = \{ B \in G : \det(B) = \det(X) \}.$$

Above proves $XS \subset \{ \quad \sim\sim\sim \quad \}$

Sonow let $B \in G$ with $\det(B) = \det(X)$.

So let's write B as

$$B = X \underline{(X^{-1}B)}$$

and note that $\det(X^{-1}B)$

$$= |X^{-1}|\,|B|$$

$$= \frac{1}{|X|}\,|X|$$

$= 1.$

$\Rightarrow X^{-1}B \in S$

So $B = X(X^{-1}B) \in XS$

Therefore $\{B : \det(B) = \det(X)\} \subset XS$

Therefore $XS = \{ \quad \}$

So the partition of $GL_n(\mathbb{R})$ by left cosets of $SL_n(\mathbb{R})$ is the partition according to value of the determinant.

Note: given any $r \in \mathbb{R}^*$

$$\left| \begin{pmatrix} r & & 0 \\ & 1 & \\ 0 & & \ddots \\ & & & 1 \end{pmatrix} \right| = r$$

Q19. Let H, K be subgroups of G

claim: $gH \cap gK$ is a left coset of $H \cap K$.

<u>Proof</u>: Given H, K are subgroup of G

we already know $H \cap K$ is a subgroup too.

Conjecture: $A = B$

$$gH \cap gK = g(H \cap K)$$

<u>Proof</u>: Let's try proving $A \subset B$ & $B \subset A$.

<u>$A \subset B$</u> So let $x \in gH \cap gK$

$\Rightarrow x \in gH$ AND $x \in gK$.

$\Rightarrow x = gh$ AND $x = gk$, for some $h \in H$ and $k \in K$.

$\Rightarrow gh = gk$

$\Rightarrow h = k$ ###

So let's write y for this element $h = k$ and note $y \in H \cap K$.

So $x = gy \in g(H \cap K)$.

Therefore $gH \cap gK \subset g(H \cap K)$.

Actually the implications all run up the page too, proving $g(H \cap K) \subset gH \cap gK$.

Hence $gH \cap gK = g(H \cap K)$.

Q13) Can we define the map

$\phi : \mathcal{L}_H \to \mathcal{R}_H$. by the rule.

$\phi(gH) = Hg$? Is this a well defined map? ~~For suppose~~ $\boxed{gH = xH}$

is it true that $Hg = Hx$?

Given any hg, for $h \in H$, can we write it as.

$hg = kx$, for some $k \in H$?

$\Rightarrow k = hgx^{-1}$

But is $hgx^{-1} \in H$?

$\Rightarrow gx^{-1} \in H$

But this won't always be so, notice it conflicts with lemma 6.3 which says.

$g^{-1}x \in H$.

Best resolution here would be to find a counter-example. G, H, x, g

with $gH = xH$, but $Hg \neq Hx$.

See example 6.2.

S_3, $K = \{(1), (12)\}$, (13), (123)

$(13)K = (123)K$

~~but~~ $K(13) \neq K(123)$